



Subtopic: Control Structures and Microprogramming

1. Control Structures in Processor Design

Definition

A **control structure** in computer architecture is the mechanism that **generates control signals** to direct the flow of data and execution of instructions within the CPU.

The **Control Unit (CU)** is responsible for:

- Fetching instructions from memory.
 - Decoding them into control signals.
 - Coordinating the operation of the ALU, registers, memory, and I/O.
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1.1 Types of Control Structures

(A) Hardwired Control

- **Concept:** Uses fixed electronic circuits (logic gates, flip-flops, counters, decoders) to directly generate control signals.
- **Characteristics:**
 - Very fast (because it is built from hardware circuits).
 - Less flexible – difficult to modify instruction set.
 - Common in RISC processors.

(B) Microprogrammed Control



- **Concept:** Control signals are generated using a sequence of **microinstructions** stored in a **control memory (microcode ROM)**.
 - **Characteristics:**
 - Easier to modify (just change microcode).
 - Slower than hardwired control (extra memory access required).
 - Common in CISC processors (e.g., Intel x86).
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1.2 Control Signals Categories

1. **Internal CPU Control Signals:** Activate ALU, select registers, etc.
 2. **Memory Control Signals:** Read, write, fetch instructions.
 3. **I/O Control Signals:** Data transfer with external devices.
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2. Microprogramming

Definition

Microprogramming is a method of implementing a control unit by storing the control signals for each machine instruction in the form of **microinstructions** in a special memory called the **Control Store**.

Each instruction in the ISA corresponds to a **microprogram**, which is a sequence of microinstructions.

2.1 Elements of Microprogramming

1. **Control Memory (CM):**



- A ROM or RAM that stores microinstructions.

2. Microinstruction:

- A binary word that specifies control signals for one clock cycle.
- Contains fields such as:
 - *Opcode field* → specifies micro-operation.
 - *Control field* → activates hardware components.
 - *Next address field* → indicates next microinstruction.

3. Microprogram:

- A sequence of microinstructions that together implement one machine-level instruction.

2.2 Types of Microinstructions

1. Horizontal Microinstructions:

- Each bit directly controls one signal.
- Very long (wide) instructions.
- Advantage: High degree of parallelism.
- Disadvantage: Requires more memory.

2. Vertical Microinstructions:

- Encoded signals → fewer bits required.
- Advantage: Compact representation.



- Disadvantage: Less parallel execution.

2.3 Advantages of Microprogramming

- Easier to **design and implement complex instructions**.
- Allows **modification or updates** by changing microcode.
- Supports **compatibility** across processor families.
- Simplifies control unit design compared to hardwired.

2.4 Disadvantages of Microprogramming

- Slower than hardwired control (extra memory access per instruction).
- Requires additional **control memory**, increasing cost.
- Not ideal for high-performance RISC processors.

✓ Summary

- **Control Structures** determine how instructions are executed in the CPU. They can be **hardwired** (fast, fixed) or **microprogrammed** (flexible, slower).
- **Microprogramming** uses stored microinstructions to implement control signals, making it easier to design complex ISAs.
- Choice between **hardwired** and **microprogrammed control** depends on processor design goals (speed vs flexibility).